

M21 2026 July 2026 N400 Component

Joanna Morris

2026-08-11

Contents

1	Setup	2
2	Load Files	3
3	N400 WORD DATA: INTERACTION WITH LP (PRIMARY ANALYSES)	4
3.1	N400 WORD DATA: BASE FREQUENCY and LP as FIXED EFFECTS; OS as COVARIATE	4
3.1.1	Model	4
3.1.2	Means and Contrasts	5
3.1.3	Plots	6
3.2	N400 WORD DATA: FAMILY SIZE and LP as FIXED EFFECTS; OS as COVARIATE	6
3.2.1	Model	6
3.2.2	Means	6
3.2.3	Plots (Interaction Not significant)	7
3.3	N400 WORD DATA: Interaction Plots— BASE FREQUENCY x LP; FAMILY SIZE x LP	7
4	N400 WORD DATA: INTERACTION WITH OS (EXPLORATORY ANALYSES)	7
4.1	N400 WORD DATA: BASE FREQUENCY and OS as FIXED EFFECTS; LP as COVARIATE	7
4.2	N400 WORD DATA: FAMILY SIZE and OS as FIXED EFFECTS; LP as COVARIATE	8
5	N400 NONWORD DATA : SYMMETRICAL MODEL, INTERACTION WITH LP AND OS	10
5.1	N400 NONWORD DATA: Family Size	10
5.1.1	Symmetric Model	10
5.1.2	Means and Contrasts	11
	Complexity × Family Size x LP Interaction	11
	Complexity × Family Size interaction averaged across LP	13
	Complexity × LP interaction averaged over Family Size:	13
	FS × LP interaction averaged over Complexity:	14
	Complexity × OS interaction averaged over Family Size:	14
5.1.3	PLOT 2-WAY INTERACTIONS	14
	FAMILY SIZE X COMPLEXITY Plot	14
	COMPLEXITY x LP Plot; FAMILY SIZE X LP Plot; COMPLEXITY x OS Plot	15
5.1.4	PLOT 3-WAY INTERACTIONS	15
	Plots: COMPLEXITY x LP faceted by Family size; FAMILY SIZE x LP Faceted by Complexity	15
	COMPLEXITY x LP and FAMILY SIZE x LP Interaction Plots	16
	Plots: COMPLEXITY x OS faceted by Family size; FAMILY SIZE x OS Faceted by Complexity	16

1 Setup

Load libraries

Set emmeans options

```
emm_options(  
  lmer.df = "kenward-roger",  
  pbkrtest.limit = 10000,  
  disable.pbkrtest = FALSE  
)
```

Set ggplot parameters

```
# Color palette  
my_palette_1 <- c(  
  "#E31A1C", # red  
  "#1F78B4" # blue  
)  
  
my_palette_2 <- c(  
  "#0072B2", # dark blue  
  "#E69F00" # orange/gold  
)  
my_linetypes_1 <- c("dashed", "solid")  
  
# Theme  
my_theme <- theme_classic(base_size = 8) +  
  theme(  
    legend.position = "bottom",  
    legend.direction = "horizontal",  
    legend.box = "horizontal"  
  )  
  
# Common guides  
my_guides <- guides(  
  colour = guide_legend(nrow = 1),  
  fill = guide_legend(nrow = 1)  
)
```

Define 2-way interaction table function

Function use for Complexity $\times$ LP table:

```
emm_complexity_LP_N400 <- emmeans(  
  m_nw_N400_fs,  
  ~ complexity | LP,  
  at = list(LP = c(-1, 0, 1))  
)  
  
compact_emm_table(  
  emm_complexity_LP_N400,  
  by_var = LP,  
  condition_prefix = "LP"  
)
```

for Family Size $\times$ LP table:

```
emm_FS_LP_N400 <- emmeans(  
  m_nw_N400_fs,  
  ~ family_size | LP,  
  at = list(LP = c(-1, 0, 1))  
)  
  
compact_emm_table(  
  emm_FS_LP_N400,  
  by_var = LP,  
  condition_prefix = "LP"  
)
```

For 3-way use this function like this:

```
emm_3way <- emmeans(  
  m_nw_N400_fs,  
  ~ complexity * family_size | LP,  
  at = list(LP = c(-1, 0, 1))  
)  
  
compact_3way_table(  
  emm_3way,  
  by_var = LP,  
  condition_prefix = "LP",  
  factor1 = complexity,  
  factor2 = family_size  
)
```

)

2 Load Files

```
# Load data: rt words
rt_w <- read_csv("rt_words_frq.csv") |>
  filter(correct == 1) |>
  mutate(
    SubjID = factor(SubjID),
    Item = factor(target),
    logRT = log(response_time),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: rt nonwords
rt_nw <- read_csv("rt_nwords_frq.csv") |>
  filter(correct == 1) |>
  mutate(
    SubjID = factor(SubjID),
    ItemID = factor(ItemID),
    complexity = factor(complexity,
      levels = c("SIMP", "COMP"),
      labels = c("Simple", "Complex")),
    logRT = log(response_time),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: erp_fs words
erp_fs_w <- read_csv("erp_fs_dmg_words.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size,
      levels = c("Small", "Large")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_fs nonwords
erp_fs_nw <- read_csv("erp_fs_dmg_nonwords.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size,
      levels = c("Small", "Large")),
    complexity = factor(complexity,
      labels = c("Simple", "Complex")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_bf words
erp_bf_w <- read_csv("erp_bf_dmg_words.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    base_frequency = factor(base_frequency,
      levels = c("Low", "High")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_bf nonwords
erp_bf_nw <- read_csv("erp_bf_dmg_nonwords.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    base_frequency = factor(base_frequency,
      levels = c("Low", "High")),
```

```

    complexity = factor(complexity,
                        labels = c("Simple", "Complex")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )
}

range(erp_fs_w$LP)

[[ [1] -2.674523  3.111265
range(erp_fs_w$OS)

[[ [1] -1.975820  1.926921

```

3 N400 WORD DATA: INTERACTION WITH LP (PRIMARY ANALYSES)

3.1 N400 WORD DATA: BASE FREQUENCY and LP as FIXED EFFECTS; OS as COVARIATE

3.1.1 Model

```

contrasts(erp_bf_w$base_frequency) <- c(-0.5, 0.5)

m_w_N400_bf <- lmer(
  value ~ base_frequency * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N400_bf)

[[ Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
[[ Formula: value ~ base_frequency * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
[[ Data: subset(erp_bf_w, time_window == "N400")
[[ Control: lmerControl(optimizer = "bobyqa")
[[
[[ REML criterion at convergence: 19468.7
[[
[[ Scaled residuals:
[[      Min       1Q   Median       3Q      Max
[[ -11.6094  -0.3522   0.0107   0.3807  10.5439
[[
[[ Random effects:
[[ Groups      Name                Variance Std.Dev.
[[ SubjID:chlabel (Intercept)  23.249    4.822
[[ SubjID       (Intercept)  18.541    4.306
[[ Residual                    9.157    3.026
[[ Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
[[
[[ Fixed effects:
[[              Estimate Std. Error    df t value Pr(>|t|)
[[ (Intercept)    5.617e-01  5.722e-01  5.700e+01  0.982  0.3304
[[ base_frequency1 2.141e-01  1.064e-01  1.618e+03  2.012  0.0444 *
[[ LP             4.677e-02  4.652e-01  5.700e+01  0.101  0.9203
[[ OS             1.290e+00  6.399e-01  5.700e+01  2.016  0.0485 *
[[ base_frequency1:LP 3.444e-01  8.659e-02  1.618e+03  3.977  7.29e-05 ***
[[ ---
[[ Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
[[
[[ Correlation of Fixed Effects:
[[              (Intr) bs_fr1 LP      OS
[[ bas_frqn1  0.000
[[ LP         0.038  0.000
[[ OS        -0.046  0.000  0.021
[[ bs_frqn1:LP 0.000  0.039  0.000  0.000
[[
[[ # Effect sizes for individual terms
eta_squared(m_w_N400_bf, partial = TRUE)

[[ # Effect Size for ANOVA (Type III)
[[
[[ Parameter      | Eta2 (partial) |      95% CI
[[ -----
[[ base_frequency |      2.50e-03 | [0.00, 1.00]

```

```

|| LP | 1.77e-04 | [0.00, 1.00]
|| OS | 0.07 | [0.00, 1.00]
|| base_frequency:LP | 9.68e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

```

# Model R²
r2_nakagawa(m_w_N400_bf)

```

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.825
|| Marginal R2: 0.026

```

3.1.2 Means and Contrasts

```

as.tibble(emmeans(m_w_N400_bf, ~ base_frequency)) |> print()

```

```

|| # A tibble: 2 x 6
|| base_frequency emmean SE df lower.CL upper.CL
|| <fct> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Low 0.515 0.574 58.0 -0.634 1.66
|| 2 High 0.712 0.574 58.0 -0.436 1.86

```

```

emm_bf_LP <- emmeans(m_w_N400_bf,
  ~ base_frequency | LP,
  at = list(LP = c(-2.5, -1, 0, 1, 3)))

```

```

emm_part <- summary(emm_bf_LP) |>
  as.data.frame() |>
  transmute(LP, Effect = paste("EMM:", base_frequency), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

```

```

contrast_part <- pairs(emm_bf_LP) |>
  summary() |>
  as.data.frame() |>
  transmute(LP, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

```

```

emm_contrast_table_LP_N400_words <- bind_rows(emm_part, contrast_part) |>
  arrange(LP, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE = round(SE, 2),
    df = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio = round(t.ratio, 2),
    p.value = round(p.value, 3)
  )

```

```

as.tibble(emm_contrast_table_LP_N400_words, digits = 3) |> print(n=20)

```

```

|| # A tibble: 15 x 9
|| LP Effect Estimate SE df lower.CL upper.CL t.ratio p.value
|| <dbl> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 -2.5 Contrast: Low - High 0.65 0.24 1618 NA NA 2.72 0.007
|| 2 -2.5 EMM: High 0.18 1.28 58 -2.39 2.74 NA NA
|| 3 -2.5 EMM: Low 0.82 1.28 58 -1.74 3.39 NA NA
|| 4 -1 Contrast: Low - High 0.13 0.13 1618 NA NA 0.97 0.333
|| 5 -1 EMM: High 0.5 0.73 58 -0.95 1.96 NA NA
|| 6 -1 EMM: Low 0.63 0.73 58 -0.82 2.09 NA NA
|| 7 0 Contrast: Low - High -0.21 0.11 1618 NA NA -2.01 0.044
|| 8 0 EMM: High 0.72 0.57 58 -0.43 1.87 NA NA
|| 9 0 EMM: Low 0.51 0.57 58 -0.64 1.66 NA NA
|| 10 1 Contrast: Low - High -0.56 0.14 1618 NA NA -3.99 0
|| 11 1 EMM: High 0.94 0.75 58 -0.57 2.45 NA NA
|| 12 1 EMM: Low 0.38 0.75 58 -1.13 1.89 NA NA
|| 13 3 Contrast: Low - High -1.25 0.28 1618 NA NA -4.38 0
|| 14 3 EMM: High 1.38 1.54 58 -1.69 4.45 NA NA
|| 15 3 EMM: Low 0.13 1.54 58 -2.94 3.21 NA NA

```

3.1.3 Plots

3.2 N400 WORD DATA: FAMILY SIZE and LP as FIXED EFFECTS; OS as COVARIATE

3.2.1 Model

```
contrasts(erp_fs_w$family_size) <- c(-0.5, 0.5)

m_w_N400_fs <- lmer(
  value ~ family_size * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N400_fs)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_w, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15561.8
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -10.9307  -0.3874   0.0081   0.4071   6.2048
||
|| Random effects:
|| Groups          Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept)  5.238      2.289
|| SubjID         (Intercept)  4.575      2.139
|| Residual                                3.304      1.818
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    2.618e-01  2.842e-01  5.700e+01  0.921  0.3608
|| family_size1    4.522e-01  6.392e-02  1.618e+03  7.074 2.23e-12 ***
|| LP              1.231e-03  2.311e-01  5.700e+01  0.005  0.9958
|| OS              6.771e-01  3.179e-01  5.700e+01  2.130  0.0375 *
|| family_size1:LP -7.114e-02  5.201e-02  1.618e+03 -1.368  0.1716
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) fmly_1 LP      OS
|| family_siz1  0.000
|| LP           0.038  0.000
|| OS          -0.046  0.000  0.021
|| fmly_sz1:LP  0.000  0.039  0.000  0.000

# Effect sizes for individual terms
eta_squared(m_w_N400_fs, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter      | Eta2 (partial) |      95% CI
|| -----|-----|-----|
|| family_size    |      0.03 | [0.02, 1.00]
|| LP             | 4.98e-07 | [0.00, 1.00]
|| OS             |      0.07 | [0.00, 1.00]
|| family_size:LP | 1.15e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

# Model R²
r2_nakagawa(m_w_N400_fs)

|| # R2 for Mixed Models
||
|| Conditional R2: 0.756
|| Marginal R2: 0.031
```

3.2.2 Means

```
as.tibble(emmeans(m_w_N400_fs, ~ family_size)) |> print()

|| # A tibble: 2 x 6
```

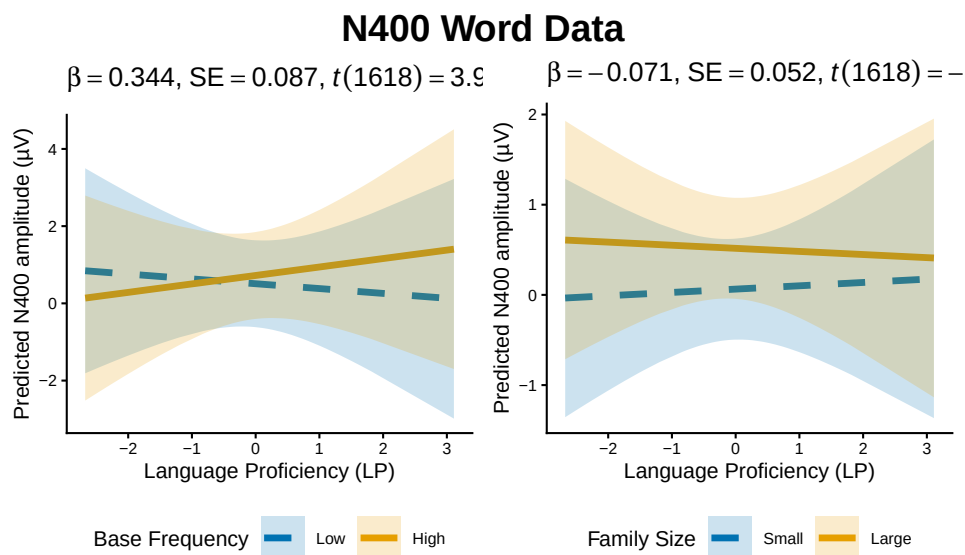
	family_size	emmean	SE	df	lower.CL	upper.CL
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Small	0.0623	0.286	58.5	-0.509	0.634
2	Large	0.518	0.286	58.5	-0.0535	1.09

3.2.3 Plots (Interaction Not significant)

3.3 N400 WORD DATA: Interaction Plots— BASE FREQUENCY x LP; FAMILY SIZE x LP

```
row1 <- plot_grid(BFxLP_n400, FSxLP_n400, ncol = 2)

plot_grid(
  ggdraw() + draw_label("N400 Word Data", fontface = "bold"),
  row1,
  ncol = 1,
  rel_heights = c(0.08, 1)
)
```



Does the slopes of the lines in the significant interactions, significantly differ from zero?

```
test(entrends(m_w_N400_bf, ~ base_frequency, var = "LP"))
```

	base_frequency	LP.trend	SE	df	t.ratio	p.value
Low	-0.125	0.467	58	-0.268	0.7893	
High	0.219	0.467	58	0.469	0.6411	

Degrees-of-freedom method: kenward-roger

4 N400 WORD DATA: INTERACTION WITH OS (EXPLORATORY ANALYSES)

4.1 N400 WORD DATA: BASE FREQUENCY and OS as FIXED EFFECTS; LP as COVARIATE

```
# BINNED BY BASE FREQUENCY
m_w_N400_bf_OS <- lmer(
  value ~ base_frequency * OS + LP +
    (1 | SubjID) + (1 | SubjID:chlabel),
  data = subset(erp_bf_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N400_bf_OS)
```

```
|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ base_frequency * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_bf_w, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
```

```

||
|| REML criterion at convergence: 19481.2
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -11.6252  -0.3573   0.0068   0.3789  10.5554
||
|| Random effects:
|| Groups      Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept) 23.212   4.818
|| SubjID       (Intercept) 18.541   4.306
|| Residual                9.232   3.038
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)   5.617e-01  5.722e-01  5.700e+01  0.982  0.3304
|| base_frequency1 1.893e-01  1.069e-01  1.618e+03  1.772  0.0767 .
|| OS            1.290e+00  6.399e-01  5.700e+01  2.016  0.0485 *
|| LP            4.677e-02  4.652e-01  5.700e+01  0.101  0.9203
|| base_frequency1:OS 1.947e-01  1.196e-01  1.618e+03  1.628  0.1036
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) bs_fr1 OS      LP
|| bas_frnc1  0.000
|| OS        -0.046  0.000
|| LP         0.038  0.000  0.021
|| bs_frnc1:OS 0.000 -0.047  0.000  0.000
||
|| # Effect sizes for individual terms
|| eta_squared(m_w_N400_bf_OS, partial = TRUE)
||
|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----|-----|-----|
|| base_frequency | 1.94e-03 | [0.00, 1.00]
|| OS | 0.07 | [0.00, 1.00]
|| LP | 1.77e-04 | [0.00, 1.00]
|| base_frequency:OS | 1.64e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
||
|| # Model R^2
|| r2_nakagawa(m_w_N400_bf_OS)
||
|| # R2 for Mixed Models
||
|| Conditional R2: 0.824
|| Marginal R2: 0.026

```

4.2 N400 WORD DATA: FAMILY SIZE and OS as FIXED EFFECTS; LP as COVARIATE

```

|| # BINNED BY FAMILY SIZE
||
|| m_w_N400_fs_OS <- lmer(
||   value ~ family_size * OS + LP +
||     (1 | SubjID) + (1 | SubjID:chlabel),
||   data = subset(erp_fs_w, time_window == "N400"),
||   control = lmerControl(optimizer = "bobyqa")
|| )
|| summary(m_w_N400_fs_OS)
||
|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_w, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15561.4
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -10.9068  -0.3881   0.0130   0.4078   6.1805
||
|| Random effects:
|| Groups      Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept) 5.237   2.289

```

```

|| SubjID          (Intercept) 4.575    2.139
|| Residual              3.304    1.818
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||               Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    2.618e-01  2.842e-01  5.700e+01   0.921   0.3608
|| family_size1   4.518e-01  6.394e-02  1.618e+03   7.066  2.37e-12 ***
|| OS             6.771e-01  3.179e-01  5.700e+01   2.130   0.0375 *
|| LP             1.231e-03  2.311e-01  5.700e+01   0.005   0.9958
|| family_size1:OS 9.090e-02  7.155e-02  1.618e+03   1.270   0.2041
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||               (Intr) fmly_1 OS      LP
|| family_size1  0.000
|| OS            -0.046  0.000
|| LP            0.038  0.000  0.021
|| fmly_sz1:OS   0.000 -0.047  0.000  0.000

```

Effect sizes for individual terms
eta_squared(m_w_N400_fs_OS, partial = TRUE)

```

|| # Effect Size for ANOVA (Type III)
||
|| Parameter      | Eta2 (partial) |      95% CI
|| -----
|| family_size    |          0.03 | [0.02, 1.00]
|| OS             |          0.07 | [0.00, 1.00]
|| LP             |       4.98e-07 | [0.00, 1.00]
|| family_size:OS |       9.97e-04 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

Model R²
r2_nakagawa(m_w_N400_fs_OS)

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.756
|| Marginal R2: 0.031

```

5 N400 NONWORD DATA : SYMMETRICAL MODEL, INTER-ACTION WITH LP AND OS

For the nonword N400, does the symmetric model we just ran replace the separate LP and OS models, or do all three get reported?***

The symmetric model is the cleanest solution — it tests both dimensions in the same model and provides the strongest evidence for the dissociation. The separate models are now superseded.

5.1 N400 NONWORD DATA: Family Size

5.1.1 Symmetric Model

```

contrasts(erp_fs_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_fs_nw$family_size) <- c(-0.5, 0.5)

m_nw_N400_fs_symmetric <- lmer(
  value ~ complexity * family_size * LP +
    complexity * family_size * OS +
    (1 | SubjID) + (1 | SubjID:chlabel),
  data = subset(erp_fs_nw, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)

summary(m_nw_N400_fs_symmetric)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * family_size * LP + complexity * family_size * OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_nw, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 32974.7
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -14.0373  -0.4804  -0.0160   0.4598  14.1912
||
|| Random effects:
|| Groups             Name             Variance Std.Dev.
|| SubjID:chlabel    (Intercept)  6.392     2.528
|| SubjID            (Intercept)  5.568     2.360
|| Residual                    6.100     2.470
|| Number of obs: 6480, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    -0.45163    0.31311    57.00007   -1.442  0.154665
|| complexity1    -0.46942    0.06147  4851.00008   -7.636  2.68e-14 ***
|| family_size1    0.08473    0.06147  4851.00008    1.378  0.168178
|| LP              0.01139    0.25458    57.00006    0.045  0.964463
|| OS              0.38123    0.35019    57.00006    1.089  0.280904
|| complexity1:family_size1 -0.44594    0.12295  4851.00008   -3.627  0.000290 ***
|| complexity1:LP    0.25409    0.04998  4851.00008    5.084  3.84e-07 ***
|| family_size1:LP    0.12220    0.04998  4851.00008    2.445  0.014528 *
|| complexity1:OS    0.23998    0.06876  4851.00008    3.490  0.000487 ***
|| family_size1:OS    0.06566    0.06876  4851.00008    0.955  0.339609
|| complexity1:family_size1:LP 0.22934    0.09997  4851.00008    2.294  0.021819 *
|| complexity1:family_size1:OS 0.14265    0.13751  4851.00008    1.037  0.299619
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) cmplx1 fmly_1 LP      OS      cm1:_1 cm1:LP f_1:LP cm1:OS f_1:OS c1:_1:L
|| complexity1  0.000
|| family_size1 0.000 0.000
|| LP           0.038 0.000 0.000
|| OS           -0.046 0.000 0.000 0.021
|| cmplxty1:_1  0.000 0.000 0.000 0.000 0.000
|| cmplxty1:LP  0.000 0.038 0.000 0.000 0.000 0.000
|| fmly_sz1:LP  0.000 0.000 0.038 0.000 0.000 0.000 0.000
|| cmplxty1:OS  0.000 -0.046 0.000 0.000 0.000 0.000 0.021 0.000
|| fmly_sz1:OS  0.000 0.000 -0.046 0.000 0.000 0.000 0.000 0.021 0.000
|| cmpl1:_1:LP  0.000 0.000 0.000 0.000 0.000 0.038 0.000 0.000 0.000
|| cmpl1:_1:OS  0.000 0.000 0.000 0.000 0.000 -0.046 0.000 0.000 0.000 0.021

# Effect Sizes
eta_squared(m_nw_N400_fs_symmetric, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||

```

```

|| Parameter | Eta2 (partial) | 95% CI
|| -----
|| complexity | 0.01 | [0.01, 1.00]
|| family_size | 3.91e-04 | [0.00, 1.00]
|| LP | 3.51e-05 | [0.00, 1.00]
|| OS | 0.02 | [0.00, 1.00]
|| complexity:family_size | 2.70e-03 | [0.00, 1.00]
|| complexity:LP | 5.30e-03 | [0.00, 1.00]
|| family_size:LP | 1.23e-03 | [0.00, 1.00]
|| complexity:OS | 2.51e-03 | [0.00, 1.00]
|| family_size:OS | 1.88e-04 | [0.00, 1.00]
|| complexity:family_size:LP | 1.08e-03 | [0.00, 1.00]
|| complexity:family_size:OS | 2.22e-04 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

```

# R-squared
r2_nakagawa(m_nw_N400_fs_symmetric)

```

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.667
|| Marginal R2: 0.013

```

5.1.2 Means and Contrasts

```

emm_options(lmer.df = "satterthwaite")

range(erp_bf_nw$LP)

```

Complexity × Family Size x LP Interaction

```

|| [1] -2.674523 3.111265

range(erp_bf_nw$OS)

```

```

|| [1] -1.975820 1.926921
# LP three-way follow-up
emm_3way_LP <- emmeans(m_nw_N400_fs_symmetric,
  ~ complexity * family_size | LP,
  at = list(LP = c(-2.5, -1, 0, 1, 3)),
  lmerTest.limit = 6480)

```

```

LP_3way_contrasts <- compact_3way_table(
  emm_3way_LP,
  by_var = LP,
  condition_prefix = "LP",
  factor1 = complexity,
  factor2 = family_size
)
as.tibble(LP_3way_contrasts) |> print(n = 50)

```

```

|| # A tibble: 45 x 9
|| Comparison Condition Estimate SE df lower.CL upper.CL t.ratio p.value
|| <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Complex Large LP = -1 -0.99 0.4 60.3 -1.79 -0.19 NA NA
|| 2 Complex Small LP = -1 -0.62 0.4 60.3 -1.42 0.18 NA NA
|| 3 Simple Large LP = -1 0.06 0.4 60.3 -0.74 0.86 NA NA
|| 4 Simple Small LP = -1 -0.24 0.4 60.3 -1.04 0.56 NA NA
|| 5 Int: Simple - Complex across Small - Large LP = -1 -0.67 0.16 4851 NA NA -4.31 0
|| 6 Small - Large LP = -1 | complexity = Complex 0.37 0.11 4851 NA NA 3.36 0.001
|| 7 Small - Large LP = -1 | complexity = Simple -0.3 0.11 4851 NA NA -2.73 0.006
|| 8 Simple - Complex LP = -1 | family_size = Large 1.05 0.11 4851 NA NA 9.55 0
|| 9 Simple - Complex LP = -1 | family_size = Small 0.38 0.11 4851 NA NA 3.45 0.001
|| 10 Complex Large LP = -2.5 -1.37 0.71 60.3 -2.79 0.04 NA NA
|| 11 Complex Small LP = -2.5 -0.65 0.71 60.3 -2.07 0.77 NA NA
|| 12 Simple Large LP = -2.5 0.23 0.71 60.3 -1.19 1.64 NA NA
|| 13 Simple Small LP = -2.5 -0.06 0.71 60.3 -1.48 1.35 NA NA
|| 14 Int: Simple - Complex across Small - Large LP = -2.5 -1.01 0.27 4851 NA NA -3.7 0
|| 15 Small - Large LP = -2.5 | complexity = Complex 0.72 0.19 4851 NA NA 3.74 0
|| 16 Small - Large LP = -2.5 | complexity = Simple -0.29 0.19 4851 NA NA -1.49 0.136
|| 17 Simple - Complex LP = -2.5 | family_size = Large 1.6 0.19 4851 NA NA 8.26 0
|| 18 Simple - Complex LP = -2.5 | family_size = Small 0.59 0.19 4851 NA NA 3.03 0.002
|| 19 Complex Large LP = 0 -0.73 0.32 60.3 -1.37 -0.1 NA NA
|| 20 Complex Small LP = 0 -0.6 0.32 60.3 -1.23 0.04 NA NA
|| 21 Simple Large LP = 0 -0.05 0.32 60.3 -0.69 0.58 NA NA
|| 22 Simple Small LP = 0 -0.36 0.32 60.3 -0.99 0.27 NA NA
|| 23 Int: Simple - Complex across Small - Large LP = 0 -0.44 0.12 4851 NA NA -3.58 0
|| 24 Small - Large LP = 0 | complexity = Complex 0.13 0.09 4851 NA NA 1.53 0.127
|| 25 Small - Large LP = 0 | complexity = Simple -0.31 0.09 4851 NA NA -3.54 0

```

```

|| 26 Simple - Complex          LP = 0 | family_size = Large      0.68 0.09 4851      NA      NA      7.82 0
|| 27 Simple - Complex          LP = 0 | family_size = Small     0.24 0.09 4851      NA      NA      2.76 0.006
|| 28 Complex Large             LP = 1                          -0.47 0.42 60.3    -1.31 0.36 NA NA
|| 29 Complex Small             LP = 1                          -0.58 0.42 60.3    -1.41 0.25 NA NA
|| 30 Simple Large              LP = 1                          -0.16 0.42 60.3    -1      0.67 NA NA
|| 31 Simple Small              LP = 1                          -0.48 0.42 60.3    -1.31 0.35 NA NA
|| 32 Int: Simple - Complex across Small - Large LP = 1          -0.21 0.16 4851      NA      NA     -1.31 0.192
|| 33 Small - Large             LP = 1 | complexity = Complex  -0.1 0.11 4851      NA      NA     -0.91 0.36
|| 34 Small - Large             LP = 1 | complexity = Simple  -0.31 0.11 4851      NA      NA     -2.76 0.006
|| 35 Simple - Complex          LP = 1 | family_size = Large  0.31 0.11 4851      NA      NA      2.72 0.007
|| 36 Simple - Complex          LP = 1 | family_size = Small  0.1 0.11 4851      NA      NA      0.88 0.381
|| 37 Complex Large             LP = 3                          0.04 0.85 60.3    -1.66 1.74 NA NA
|| 38 Complex Small             LP = 3                         -0.54 0.85 60.3    -2.24 1.16 NA NA
|| 39 Simple Large              LP = 3                         -0.39 0.85 60.3    -2.09 1.31 NA NA
|| 40 Simple Small              LP = 3                         -0.72 0.85 60.3    -2.42 0.98 NA NA
|| 41 Int: Simple - Complex across Small - Large LP = 3          0.25 0.33 4851      NA      NA      0.76 0.45
|| 42 Small - Large             LP = 3 | complexity = Complex  -0.58 0.23 4851      NA      NA     -2.49 0.013
|| 43 Small - Large             LP = 3 | complexity = Simple  -0.33 0.23 4851      NA      NA     -1.42 0.155
|| 44 Simple - Complex          LP = 3 | family_size = Large  -0.43 0.23 4851      NA      NA     -1.84 0.066
|| 45 Simple - Complex          LP = 3 | family_size = Small  -0.18 0.23 4851      NA      NA     -0.77 0.441

```

```

# OS three-way follow-up
emm_3way_OS <- emmeans(m_nw_N400_fs_symmetric,
  ~ family_size * complexity | OS,
  at = list(OS = c(-1.9, -1, 0, 1, 1.9)),
  lmerTest.limit = 6480)

```

```

OS_3way_contrasts <- compact_3way_table(
  emm_3way_OS,
  by_var = OS,
  condition_prefix = "OS",
  factor1 = complexity,
  factor2 = family_size
)
as.tibble(OS_3way_contrasts) |> print(n = 50)

```

```

|| # A tibble: 45 x 9
||   Comparison          Condition Estimate SE    df lower.CL upper.CL t.ratio p.value
||   <chr>              <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Complex Large      OS = -1      -1.34 0.49 60.3    -2.31 -0.36 NA NA
|| 2 Complex Small      OS = -1      -1.05 0.49 60.3    -2.03 -0.08 NA NA
|| 3 Simple Large        OS = -1      -0.32 0.49 60.3    -1.29 0.66 NA NA
|| 4 Simple Small        OS = -1      -0.63 0.49 60.3    -1.6 0.35 NA NA
|| 5 Int: Simple - Complex across Small - Large OS = -1      -0.6 0.19 4851      NA NA     -3.18 0.001
|| 6 Small - Large      OS = -1 | complexity = Complex  0.29 0.13 4851      NA NA      2.15 0.032
|| 7 Small - Large      OS = -1 | complexity = Simple  -0.31 0.13 4851      NA NA     -2.35 0.019
|| 8 Simple - Complex    OS = -1 | family_size = Large  1.02 0.13 4851      NA NA      7.66 0
|| 9 Simple - Complex    OS = -1 | family_size = Small  0.42 0.13 4851      NA NA      3.16 0.002
|| 10 Complex Large      OS = -1.9     -1.85 0.76 60.3    -3.37 -0.33 NA NA
|| 11 Complex Small      OS = -1.9     -1.44 0.76 60.3    -2.96 0.08 NA NA
|| 12 Simple Large        OS = -1.9     -0.55 0.76 60.3    -2.07 0.97 NA NA
|| 13 Simple Small        OS = -1.9     -0.87 0.76 60.3    -2.39 0.65 NA NA
|| 14 Int: Simple - Complex across Small - Large OS = -1.9     -0.73 0.29 4851      NA NA     -2.48 0.013
|| 15 Small - Large      OS = -1.9 | complexity = Complex  0.41 0.21 4851      NA NA      1.97 0.049
|| 16 Small - Large      OS = -1.9 | complexity = Simple  -0.32 0.21 4851      NA NA     -1.53 0.126
|| 17 Simple - Complex    OS = -1.9 | family_size = Large  1.3 0.21 4851      NA NA      6.12 0
|| 18 Simple - Complex    OS = -1.9 | family_size = Small  0.57 0.21 4851      NA NA      2.76 0.006
|| 19 Complex Large      OS = 0       -0.77 0.32 60.3    -1.4 -0.13 NA NA
|| 20 Complex Small      OS = 0       -0.62 0.32 60.3    -1.25 0.02 NA NA
|| 21 Simple Large        OS = 0       -0.06 0.32 60.3    -0.69 0.58 NA NA
|| 22 Simple Small        OS = 0       -0.37 0.32 60.3    -1 0.27 NA NA
|| 23 Int: Simple - Complex across Small - Large OS = 0       -0.46 0.12 4851      NA NA     -3.72 0
|| 24 Small - Large      OS = 0 | complexity = Complex  0.15 0.09 4851      NA NA      1.72 0.085
|| 25 Small - Large      OS = 0 | complexity = Simple  -0.31 0.09 4851      NA NA     -3.54 0
|| 26 Simple - Complex    OS = 0 | family_size = Large  0.71 0.09 4851      NA NA      8.17 0
|| 27 Simple - Complex    OS = 0 | family_size = Small  0.25 0.09 4851      NA NA      2.91 0.004
|| 28 Complex Large      OS = 1       -0.2 0.47 60.3    -1.13 0.73 NA NA
|| 29 Complex Small      OS = 1       -0.19 0.47 60.3    -1.12 0.74 NA NA
|| 30 Simple Large        OS = 1        0.2 0.47 60.3    -0.73 1.13 NA NA
|| 31 Simple Small        OS = 1       -0.1 0.47 60.3    -1.03 0.83 NA NA
|| 32 Int: Simple - Complex across Small - Large OS = 1       -0.31 0.18 4851      NA NA     -1.75 0.081
|| 33 Small - Large      OS = 1 | complexity = Complex  0.01 0.13 4851      NA NA      0.1 0.921
|| 34 Small - Large      OS = 1 | complexity = Simple  -0.3 0.13 4851      NA NA     -2.37 0.018
|| 35 Simple - Complex    OS = 1 | family_size = Large  0.4 0.13 4851      NA NA      3.13 0.002
|| 36 Simple - Complex    OS = 1 | family_size = Small  0.08 0.13 4851      NA NA      0.66 0.507
|| 37 Complex Large      OS = 1.9      0.31 0.73 60.3    -1.15 1.78 NA NA
|| 38 Complex Small      OS = 1.9      0.2 0.73 60.3    -1.26 1.67 NA NA
|| 39 Simple Large        OS = 1.9      0.43 0.73 60.3    -1.03 1.9 NA NA
|| 40 Simple Small        OS = 1.9      0.14 0.73 60.3    -1.33 1.6 NA NA
|| 41 Int: Simple - Complex across Small - Large OS = 1.9     -0.19 0.28 4851      NA NA     -0.66 0.512
|| 42 Small - Large      OS = 1.9 | complexity = Complex  -0.11 0.2 4851      NA NA     -0.55 0.581

```

43 Small - Large	OS = 1.9 complexity = Simple	-0.3	0.2	4851	NA	NA	-1.48	0.139
44 Simple - Complex	OS = 1.9 family_size = Large	0.12	0.2	4851	NA	NA	0.59	0.554
45 Simple - Complex	OS = 1.9 family_size = Small	-0.07	0.2	4851	NA	NA	-0.34	0.737

```
emm_complexity_FS_N400 <- emmeans(
  m_nw_N400_fs_symmetric,
  ~ complexity | family_size,
  lmerTest.limit = 6480)

complexityxfamily_size_table <- as.tibble(compact_emm_table(
  emm_complexity_FS_N400,
  by_var = family_size
)) |> print()
```

Complexity × Family Size interaction averaged across LP

# A tibble: 6 x 9	Comparison	Condition	Estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 Complex	Large	-0.74	0.32	60.3	-1.38	-0.11	NA	NA	
2 Simple	Large	-0.05	0.32	60.3	-0.68	0.59	NA	NA	
3 Simple - Complex	Large	0.7	0.09	4851	NA	NA	8.03	0	
4 Complex	Small	-0.6	0.32	60.3	-1.23	0.03	NA	NA	
5 Simple	Small	-0.35	0.32	60.3	-0.99	0.28	NA	NA	
6 Simple - Complex	Small	0.25	0.09	4851	NA	NA	2.84	0.005	

```
emm_complexity_CMP_N400 <- emmeans(
  m_nw_N400_fs_symmetric,
  ~ family_size | complexity,
  lmerTest.limit = 6480)

family_size_x_complexity_table <- as.tibble(compact_emm_table(
  emm_complexity_CMP_N400,
  by_var = complexity
)) |> print()
```

# A tibble: 6 x 9	Comparison	Condition	Estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 Large	Complex	-0.74	0.32	60.3	-1.38	-0.11	NA	NA	
2 Small	Complex	-0.6	0.32	60.3	-1.23	0.03	NA	NA	
3 Small - Large	Complex	0.14	0.09	4851	NA	NA	1.66	0.097	
4 Large	Simple	-0.05	0.32	60.3	-0.68	0.59	NA	NA	
5 Small	Simple	-0.35	0.32	60.3	-0.99	0.28	NA	NA	
6 Small - Large	Simple	-0.31	0.09	4851	NA	NA	-3.54	0	

transmute() creates a new data frame containing only the specified computations. factor() is used to encode a vector as a factor

```
emm_complexity_LP_N400 <- emmeans(
  m_nw_N400_fs_symmetric,
  ~ complexity | LP,
  at = list(LP = c(-2.5, -1, 0, 1, 3)),
  lmerTest.limit = 6480
)
```

```
complexityxLP_table <- as.tibble(compact_emm_table(
  emm_complexity_LP_N400,
  by_var = LP,
  condition_prefix = "LP"
)) |> print(n = 20)
```

Complexity × LP interaction averaged over Family Size:

# A tibble: 15 x 9	Comparison	Condition	Estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 Complex	LP = -1	-0.8	0.4	58.1	-1.6	-0.01	NA	NA	
2 Simple	LP = -1	-0.09	0.4	58.1	-0.89	0.71	NA	NA	
3 Simple - Complex	LP = -1	0.71	0.08	4851	NA	NA	9.19	0	
4 Complex	LP = -2.5	-1.01	0.7	58.1	-2.42	0.39	NA	NA	
5 Simple	LP = -2.5	0.08	0.7	58.1	-1.32	1.49	NA	NA	
6 Simple - Complex	LP = -2.5	1.09	0.14	4851	NA	NA	7.99	0	
7 Complex	LP = 0	-0.67	0.31	58.1	-1.29	-0.04	NA	NA	
8 Simple	LP = 0	-0.21	0.31	58.1	-0.84	0.42	NA	NA	
9 Simple - Complex	LP = 0	0.46	0.06	4851	NA	NA	7.48	0	
10 Complex	LP = 1	-0.53	0.41	58.1	-1.35	0.3	NA	NA	
11 Simple	LP = 1	-0.32	0.41	58.1	-1.15	0.5	NA	NA	
12 Simple - Complex	LP = 1	0.21	0.08	4851	NA	NA	2.54	0.011	
13 Complex	LP = 3	-0.25	0.84	58.1	-1.93	1.43	NA	NA	
14 Simple	LP = 3	-0.55	0.84	58.1	-2.24	1.13	NA	NA	

```
|| 15 Simple - Complex LP = 3          -0.3  0.16 4851      NA      NA      -1.84  0.065
```

```
emm_FS_LP_N400 <- emmeans(
  m_nw_N400_fs_symmetric,
  ~ family_size | LP,
  at = list(LP = c(-2.5, -1, 0, 1, 3)),
  lmerTest.limit = 6480
)

family_size_x_LP_table <- as.tibble(compact_emm_table(
  emm_FS_LP_N400,
  by_var = LP,
  condition_prefix = "LP"
)) |> print()
```

FS × LP interaction averaged over Complexity:

```
|| # A tibble: 15 x 9
||   Comparison      Condition Estimate      SE      df lower.CL upper.CL t.ratio p.value
||   <chr>          <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Large          LP = -1      -0.46  0.4   58.1   -1.26  0.33    NA      NA
|| 2 Small          LP = -1      -0.43  0.4   58.1   -1.23  0.37    NA      NA
|| 3 Small - Large LP = -1        0.03  0.08 4851    NA      NA      0.45  0.655
|| 4 Large          LP = -2.5     -0.57  0.7   58.1   -1.98  0.83    NA      NA
|| 5 Small          LP = -2.5     -0.36  0.7   58.1   -1.76  1.05    NA      NA
|| 6 Small - Large LP = -2.5     0.22  0.14 4851    NA      NA      1.59  0.112
|| 7 Large          LP = 0        -0.39  0.31  58.1   -1.02  0.24    NA      NA
|| 8 Small          LP = 0        -0.48  0.31  58.1   -1.11  0.15    NA      NA
|| 9 Small - Large LP = 0        -0.09  0.06 4851    NA      NA     -1.42  0.154
|| 10 Large         LP = 1        -0.32  0.41  58.1   -1.15  0.51    NA      NA
|| 11 Small         LP = 1        -0.53  0.41  58.1   -1.36  0.3     NA      NA
|| 12 Small - Large LP = 1        -0.21  0.08 4851    NA      NA     -2.6   0.009
|| 13 Large         LP = 3        -0.17  0.84  58.1   -1.86  1.51    NA      NA
|| 14 Small         LP = 3        -0.63  0.84  58.1   -2.31  1.05    NA      NA
|| 15 Small - Large LP = 3        -0.45  0.16 4851    NA      NA     -2.76  0.006
```

```
emm_complexity_OS_N400 <- emmeans(
  m_nw_N400_fs_symmetric,
  ~ complexity | OS,
  at = list(OS = c(-1.9, -1, 0, 1, 1.9)),
  lmerTest.limit = 6480
)

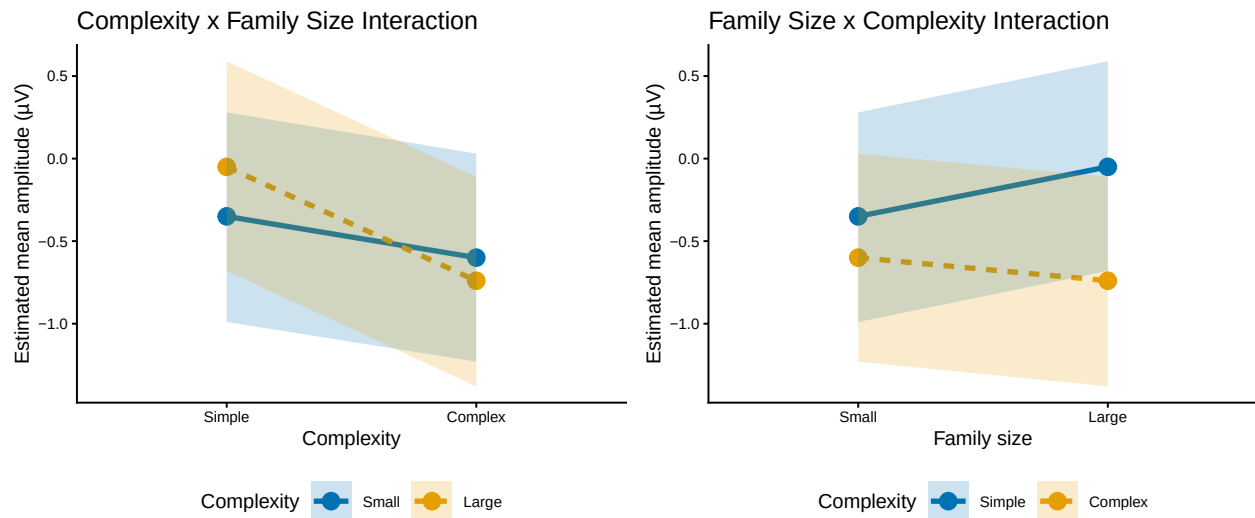
complexity_x_OS_table <- as.tibble(compact_emm_table(
  emm_complexity_OS_N400,
  by_var = OS,
  condition_prefix = "OS"
)) |> print(n = 20)
```

Complexity × OS interaction averaged over Family Size:

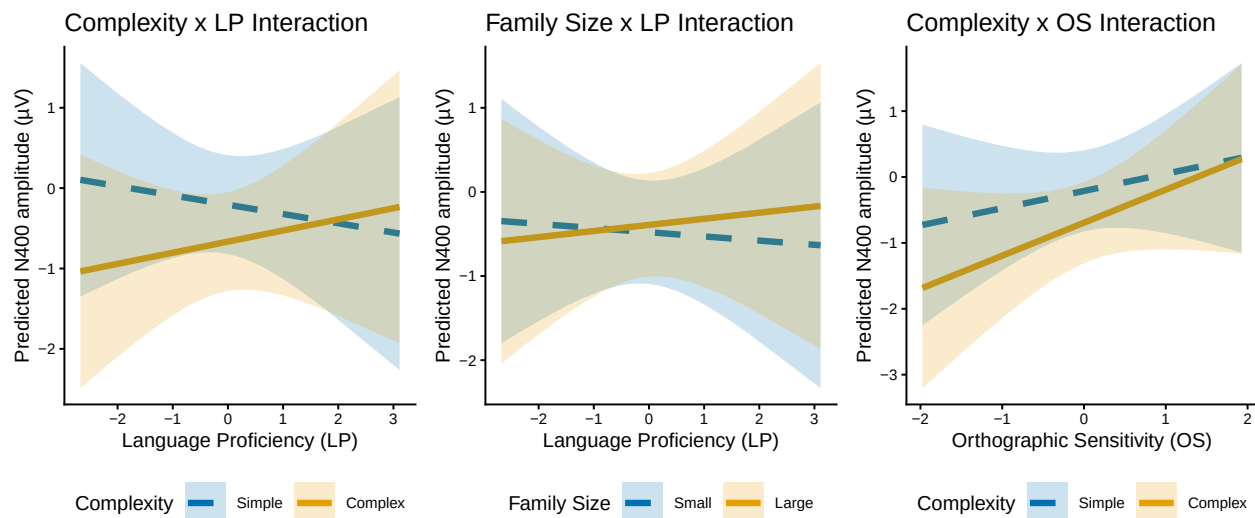
```
|| # A tibble: 15 x 9
||   Comparison      Condition Estimate      SE      df lower.CL upper.CL t.ratio p.value
||   <chr>          <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Complex          OS = -1      -1.19  0.48  58.1   -2.16 -0.23    NA      NA
|| 2 Simple           OS = -1      -0.47  0.48  58.1   -1.44  0.49    NA      NA
|| 3 Simple - Complex OS = -1        0.72  0.09 4851    NA      NA      7.65    0
|| 4 Complex          OS = -1.9     -1.65  0.75  58.1   -3.15 -0.14    NA      NA
|| 5 Simple           OS = -1.9     -0.71  0.75  58.1   -2.21  0.8     NA      NA
|| 6 Simple - Complex OS = -1.9     0.94  0.15 4851    NA      NA      6.38    0
|| 7 Complex          OS = 0        -0.69  0.31  58.1   -1.32 -0.06    NA      NA
|| 8 Simple           OS = 0        -0.21  0.31  58.1   -0.84  0.42    NA      NA
|| 9 Simple - Complex OS = 0        0.48  0.06 4851    NA      NA      7.84    0
|| 10 Complex         OS = 1        -0.19  0.46  58.1   -1.11  0.73    NA      NA
|| 11 Simple          OS = 1        0.05  0.46  58.1   -0.87  0.97    NA      NA
|| 12 Simple - Complex OS = 1        0.24  0.09 4851    NA      NA      2.68  0.007
|| 13 Complex         OS = 1.9     0.26  0.73  58.1   -1.19  1.71    NA      NA
|| 14 Simple          OS = 1.9     0.29  0.73  58.1   -1.17  1.74    NA      NA
|| 15 Simple - Complex OS = 1.9     0.03  0.14 4851    NA      NA      0.18  0.856
```

5.1.3 PLOT 2-WAY INTERACTIONS

FAMILY SIZE X COMPLEXITY Plot

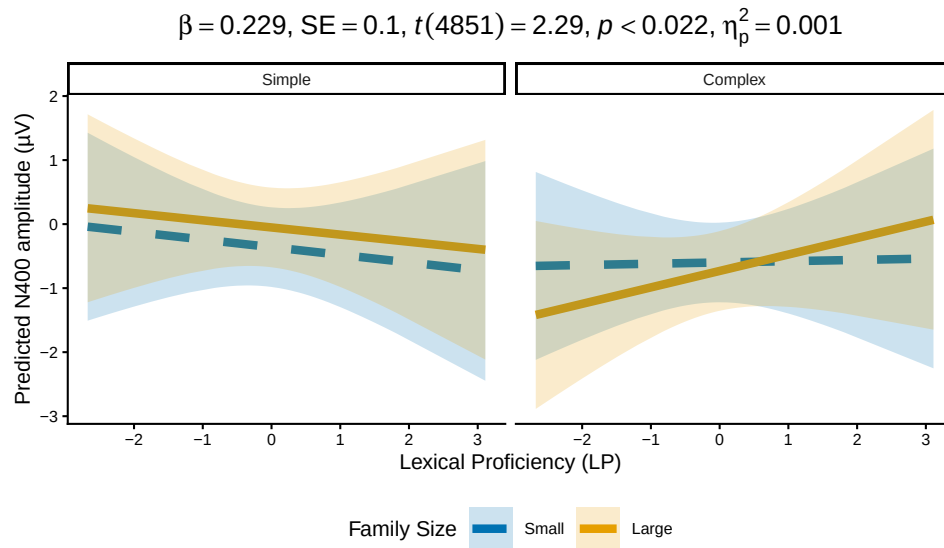


COMPLEXITY x LP Plot; FAMILY SIZE X LP Plot; COMPLEXITY x OS Plot



5.1.4 PLOT 3-WAY INTERACTIONS

Plots: COMPLEXITY x LP faceted by Family size; FAMILY SIZE x LP Faceted by Complexity

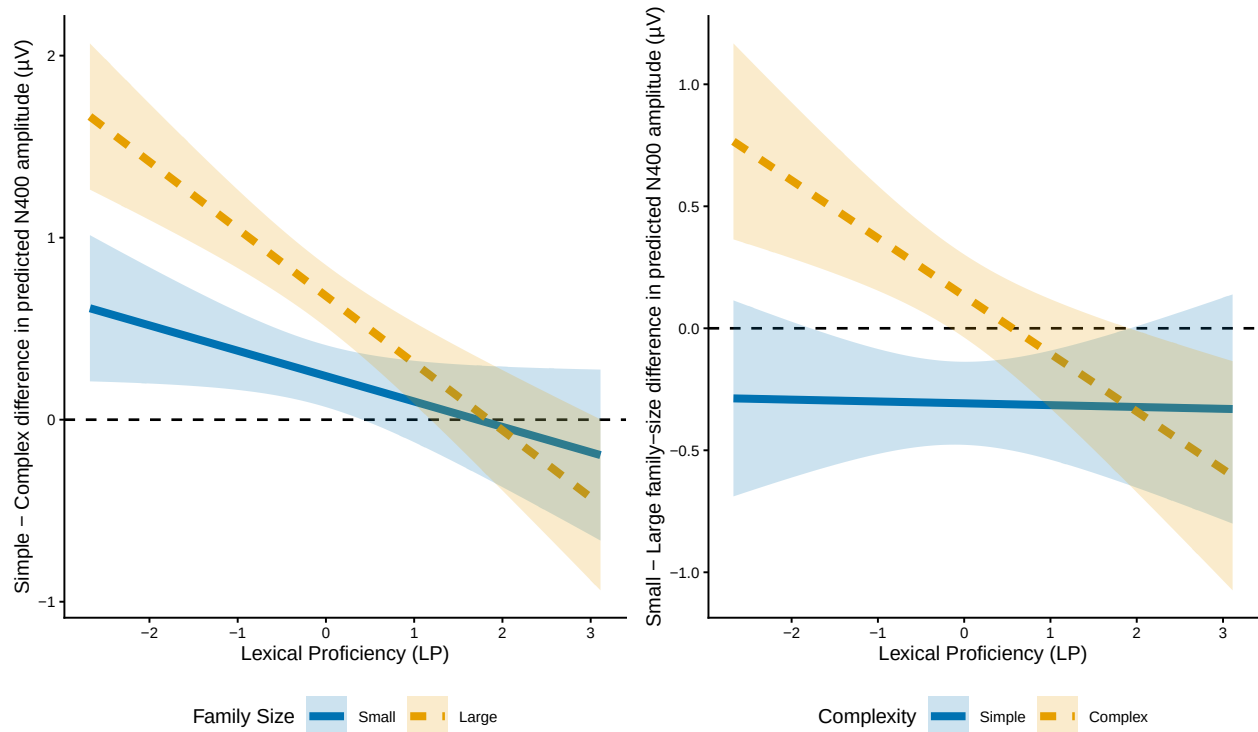


COMPLEXITY x LP and FAMILY SIZE x LP Interaction Plots Plot Simple-Complex difference across LP

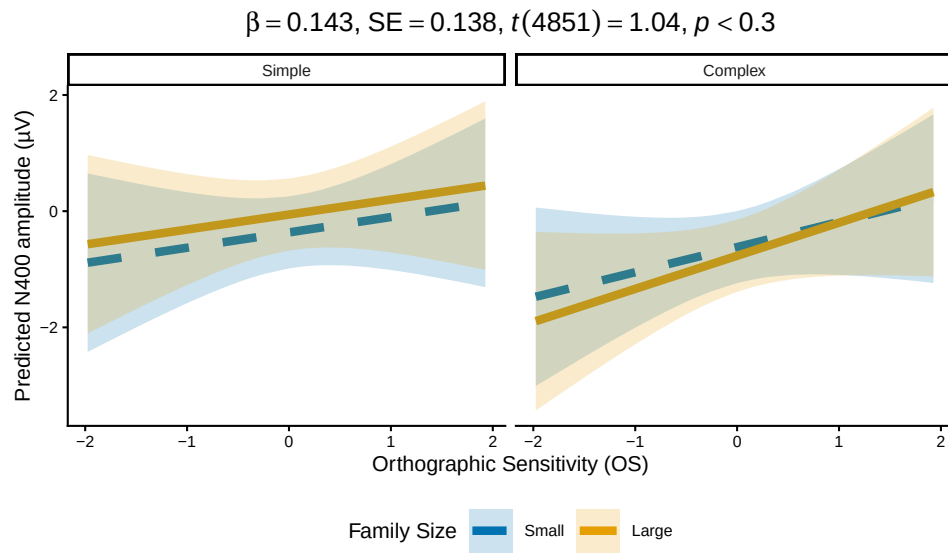
Plot Small Family - Large Family difference across LP

Print Both Plots

N400 NonWord Data



Plots: COMPLEXITY x OS faceted by Family size; FAMILY SIZE x OS Faceted by Complexity



Plot Small Family - Large Family difference across OS